



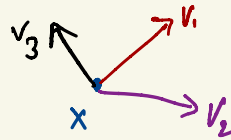
What Multi-head Attention Does?

A standard multi-head attention block has h heads, where each head h does this:

$$\text{head}^{(h)} = \text{Attn}(XW_Q^{(h)}, XW_K^{(h)}, XW_V^{(h)})$$

where each head learns $W_Q^{(h)}, W_K^{(h)}, W_V^{(h)}$.

So with H heads, the model learns H different "views" of the same input. This can be viewed as multiple vectors acting on a given input.



However, vector algebra says

many useful views of a vector can be generated from the basis vectors applying structured transformations.

Can we then learn a more efficient Attention mechanism utilizing basis vectors and their transformations without learning 8 to 16 unrelated heads?

This is the core IDEA of this work.

Let's start simple:

Let a token be represented by a vector $x \in \mathbb{R}^d$. A single scalar measurement of this vector along some other vector $u \in \mathbb{R}^d$ is simply the dot product:

$$y = u^T x \quad (\text{Tells us value of } x \text{ in direction of } u)$$

If we use many different vectors in different directions, we get $u_1^T x, u_2^T x, \dots, u_H^T x$.

i.e., we get the scaled projection of input vector x in many different directions.

This is in line with the spirit of multi-head attention, where each head asks a different question about the same token representation.

Basically, in standard multi-head attention, the different heads are aggregated by passing concatenated multi-head tokens through

MLP. This is given as $y_i = [y_i^{(1)} \parallel y_i^{(2)} \parallel \dots \parallel y_i^{(H)}] W_0$

which is a learned linear transform.

In this work, we argue that $u_1, u_2, \dots, u_n$ don't have to be learned independently. They can be generated from one base vector $u \in \mathbb{R}^d$ by structured transformations:

$$u_h = T_h u.$$

The measurements then become $u_h^T x = (T_h u)^T x$.

So instead of learning many independent vectors, we can leverage structured transformations to learn efficient multi-head attention.

A standard attention head is a learned bilinear comparison:

for one attention head, we compute

$$q_i = W_Q x_i$$

$$k_j = W_K x_j$$

we then compare the two tokens by $q_i^T k_j$.

$$\Rightarrow q_i^T k_j = x_i^T W_Q^T W_K x_j$$

$$= x_i^T B x_j \quad (\text{where } B := W_Q^T W_K)$$

The attention score is thus given as $\alpha_i^T B \alpha_j$. This is the Bilinear form.

At this scoring level, each head defines a learned bilinear geometry :

$$\text{head } h \iff B_h = W_Q^{(h)T} W_K^{(h)}.$$

Multi-head attention learns many such bilinear forms $B_1, B_2, \dots, B_H$, each one defines a different way to say how related token i and j are.

In essence, using basic vector-algebra terms, a multi-head attention is a collection of learned comparison geometries.

Avoid learning many unrelated geometries :

Standard multi-head attention basically says the different

Bilinear scoring matrices $B_1, B_2, \dots, B_H$ are all independent.

Our idea says they are related and can be generated by

known or Learned transformations of Basis vectors.

Given a learned bilinear scoring matrix B_0 , we define any other bilinear scoring matrix B_h as a symmetric transformation of the base B_0 :

$$B_h := T_h^Q{}^T B_0 T_h^K.$$

We now do not learn head specific $W_Q^{(h)}$ and $W_K^{(h)}$ matrices. Instead, each head learns shared projections

$$q_i = W_Q x_i, \quad k_j = W_K x_j,$$

and a head specific transformation metric M_h . The new scoring function is now given as :

$$\begin{aligned} \alpha_{ij}^{(h)} &= (W_Q x_i)^T M_h (W_K x_j) \\ &= x_i^T (W_Q^T M_h W_K) x_j \end{aligned}$$

$\therefore$ The bilinear scoring matrix is now given as

$$B_h = W_Q^T M_h W_K \quad \text{--- ①}$$

Intuitively, this means that the proposed bilinear scoring matrix is a factorization of the standard head-specific bilinear form.

Is Eq ① really a symmetric transformation of B_0 ?

Suppose each head is generated by transforming the shared query and key:

$$q_i^{(h)} = T_h^Q q_i \quad ; \quad k_j^{(h)} = T_h^K q_j.$$

Then the head score is:

$$\alpha_{ij}^{(h)} = q_i^{(h)T} k_j^{(h)} = q_i^T (T_h^Q)^T T_h^K q_j$$

So in this case, $M_h = (T_h^Q)^T T_h^K$.

$\therefore$ In this form, the head score is given by

$$\alpha_{ij}^{(h)} = x_i^T W_Q^T (T_h^Q)^T T_h^K W_K x_j$$

where $B_h = W_Q^T (T_h^Q)^T T_h^K W_K$

Well, what do we get out of this new formulation?

1) We don't learn H query $W_Q^{(h)}$ and key $W_K^{(h)}$ matrices

anymore, rather we learn shared W_Q and W_K and H

transformation matrices M_h . This cuts down the number

of learned parameters by a lot.

2) More importantly, the multiple heads are not trying to learn the scoring geometry independently from scratch. they rather learn to look at the input vector from different views using structured transformation of the base query and key matrices. This helps the sample efficiency by a lot.

In short, we have :

many heads $\approx$ one base representation
+ many transformed matrices.

The Geometric Interpretation: heads are different coordinates

A vector itself is abstract. coordinates depend on the basis used to describe it. For instance, if we rotate, scale, reflect, shear or transform a coordinate system, the same vector can reveal different structure.

In linear algebra, a change of basis looks like: $x \mapsto Tx$
where T is a transformation matrix.

The attention score in a transformed coordinate system becomes

$$\begin{aligned}\alpha_{ij} &= (T^Q q_i)^T (T^K k_j) \\ &= q_i^T (T^Q)^T T^K k_j\end{aligned}$$

where $M = (T^Q)^T T^K$.

Hence, the attention score in the transformed frame is now :

$$\alpha_{ij} = q_i^T M k_j.$$

So each transformed head is equivalent to using a different matrix M .

In other words, multi-head attention should not require independent projected vector spaces. Rather, it may be enough to learn one vector space and several structured transformations of its metric.

Group or Lie Group perspective:

If T_1 and T_2 are transformations, then $T_2 \cdot T_1$ is also a transformation. The set of invertible linear transformations form a group $GL(d)$.

$GL(d)$ has many useful transformation subgroups such as $SO(d)$, $O(d)$, $SE(3)$ etc. A Lie group gives continuous transformation family.

Instead of learning a separate transformation M_h for each head, we instead learn a few generators:

$$A = \{A_1, A_2, \dots, A_m\}$$

A head transformation can then be generated as

$$\begin{aligned} T_h &= \exp(\theta_h A) \\ &= \exp(\theta_{h1} A_1 + \theta_{h2} A_2 + \dots + \theta_{hm} A_m) \end{aligned}$$

So head h is not an independent projection. It is a point on a learned transformation manifold.

Hence, heads are nothing but samples from a low-dimensional transformation group.

Overall, you learn 1 or 2 base vector spaces, m generators, and H small head coordinates.

An essential condition to consider:

All elements of a group are orthogonal, i.e., $\bar{A}^T A = I$.
Hence, if in our new scoring function $\alpha_{ij} = q^T (T^Q)^T (T^K) k$,
 $T^Q = T^K$, the attention score does not change. So rotating both query and key by the same same orthogonal transformation gives no new attention pattern.

It is crucial to make sure $T^Q \neq T^K$. To avoid this issue completely, we directly learn the transformation M_h as

$$M_h = \exp\left(\sum_{r=1}^m \theta_{hr} A_r\right).$$